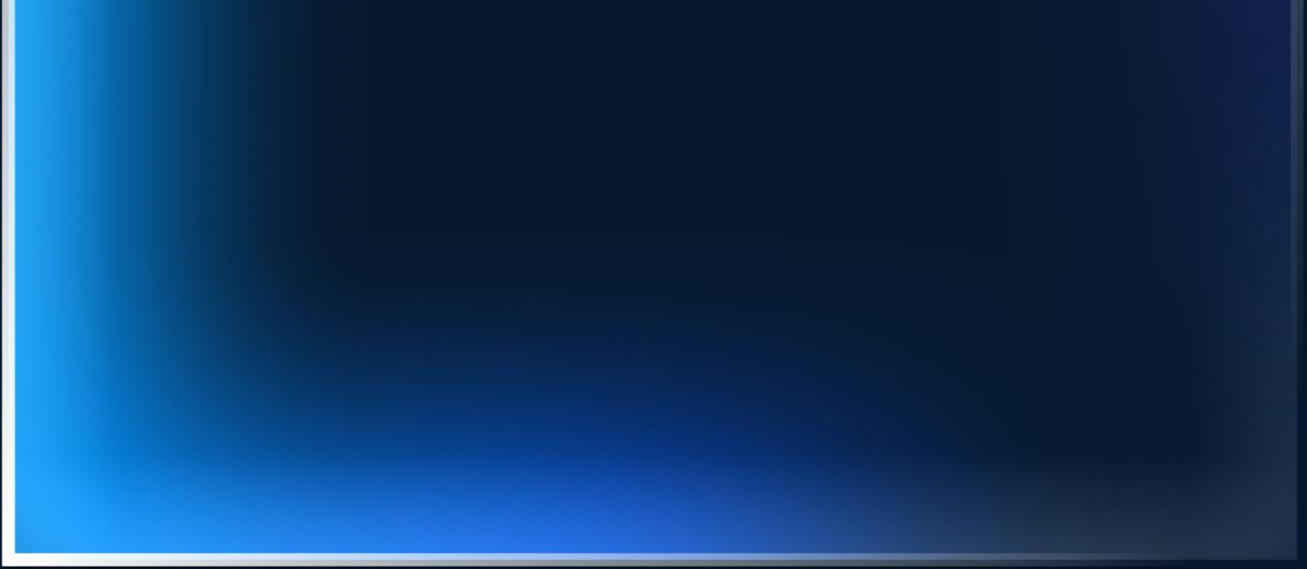


SRv6 Supercharged

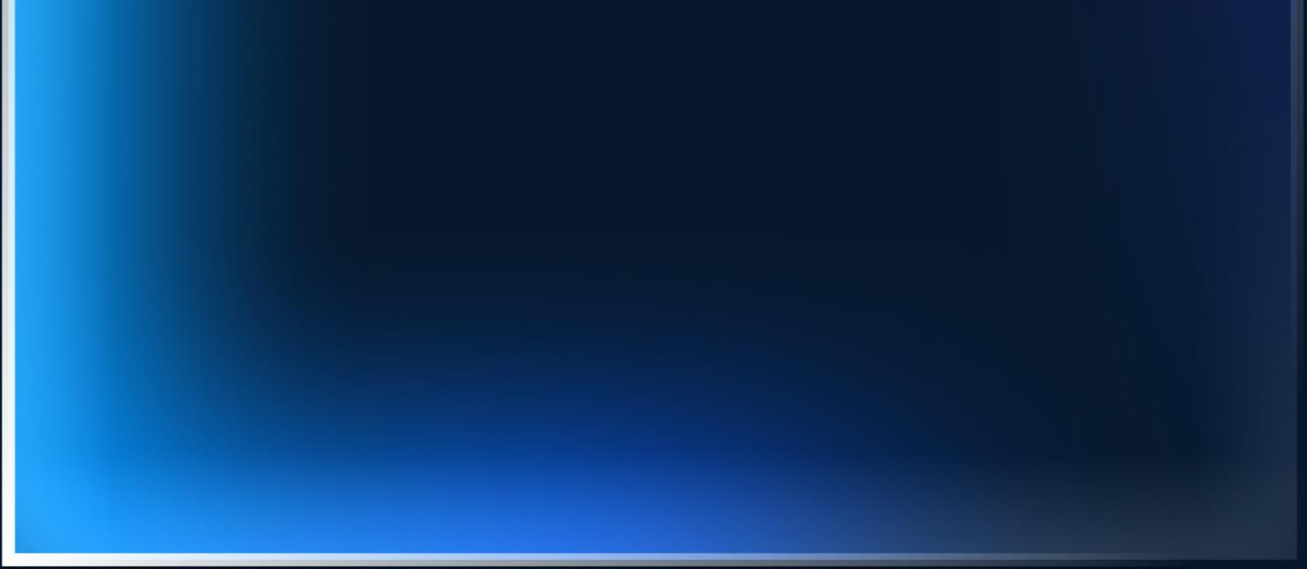
Kedar Karmarkar
Principal Engineer, Technical Marketing
Cisco **8000** Series **Secure** Routers
Enterprise Routing

23/09/2026



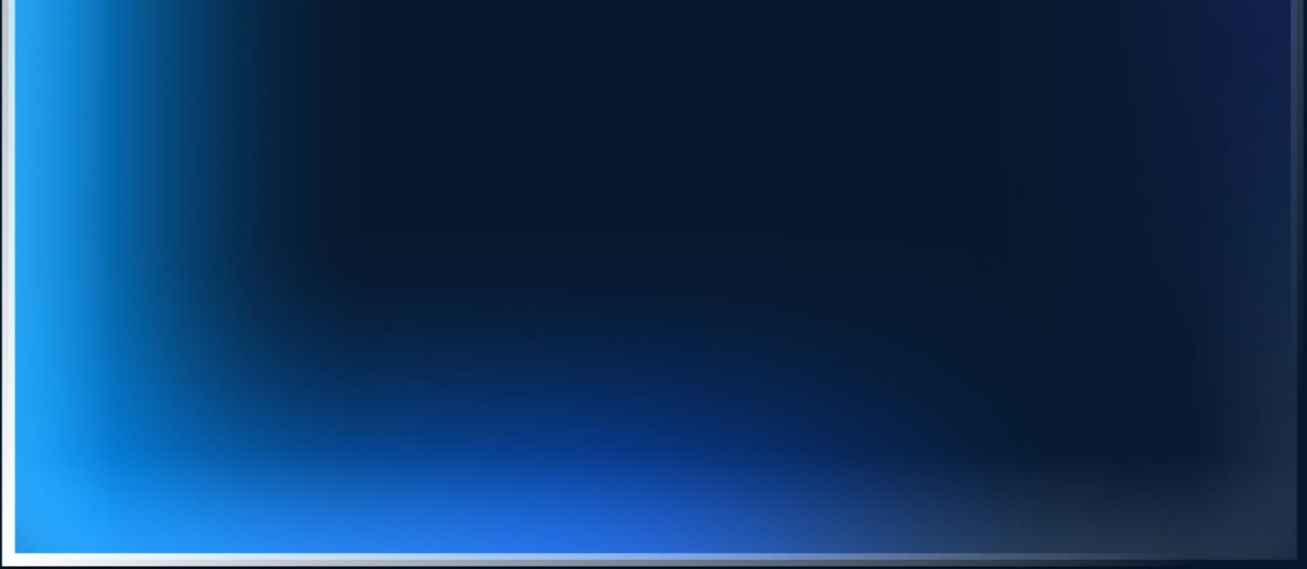
IF YOU CAN DEPLOY ...





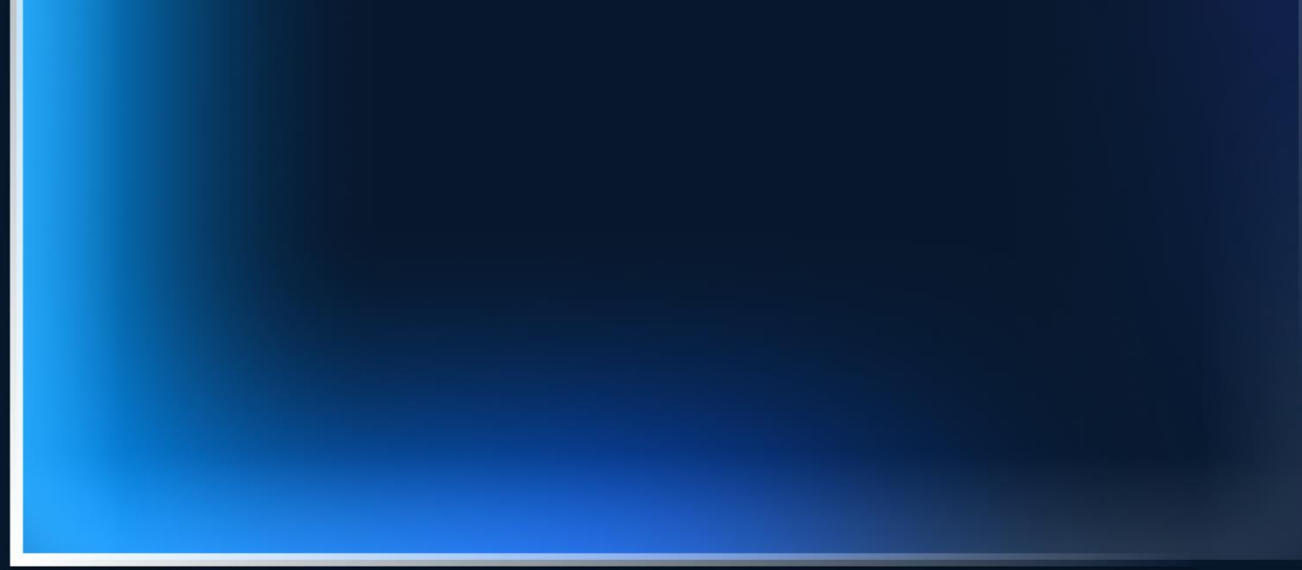
Application-Aware Routing





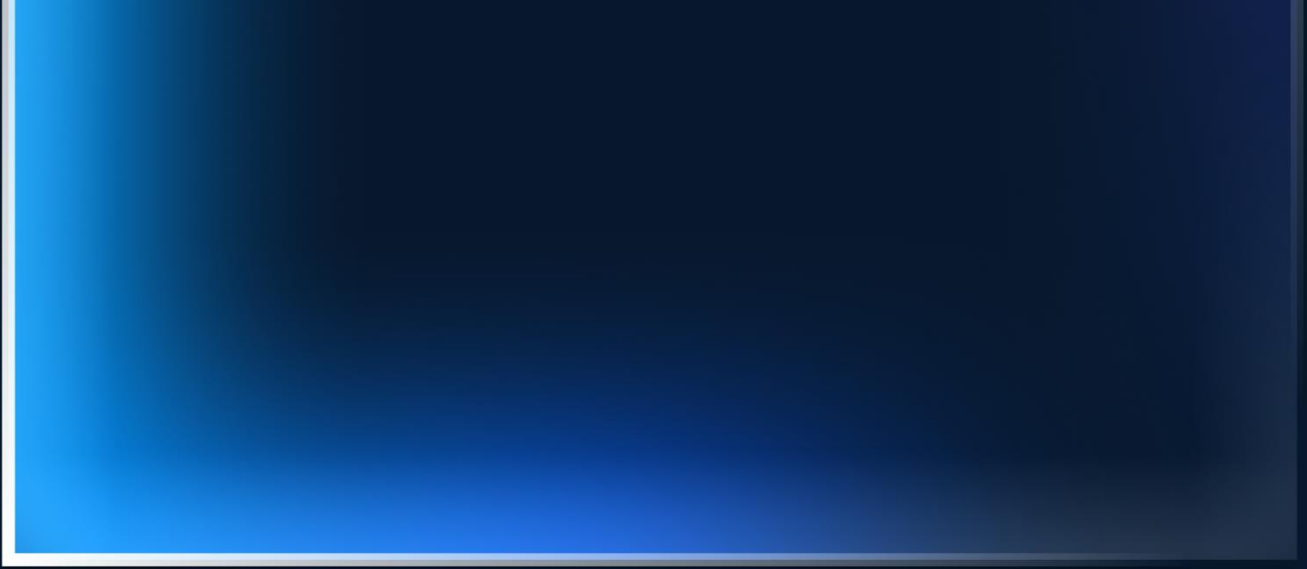
Advanced Firewalling





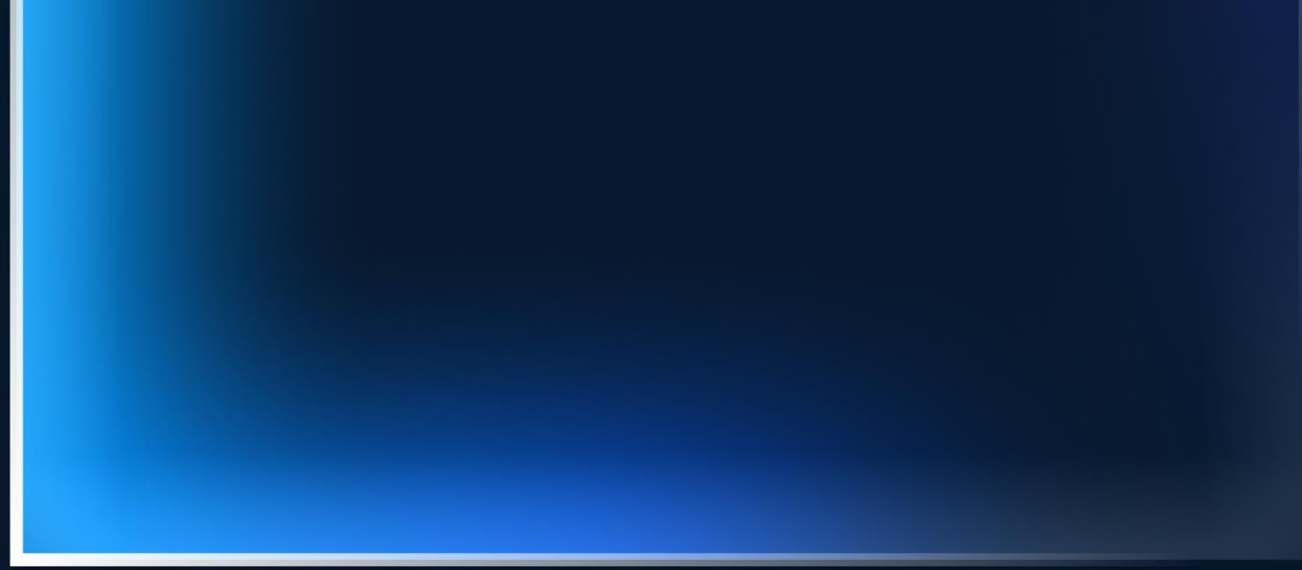
Network Address Translation





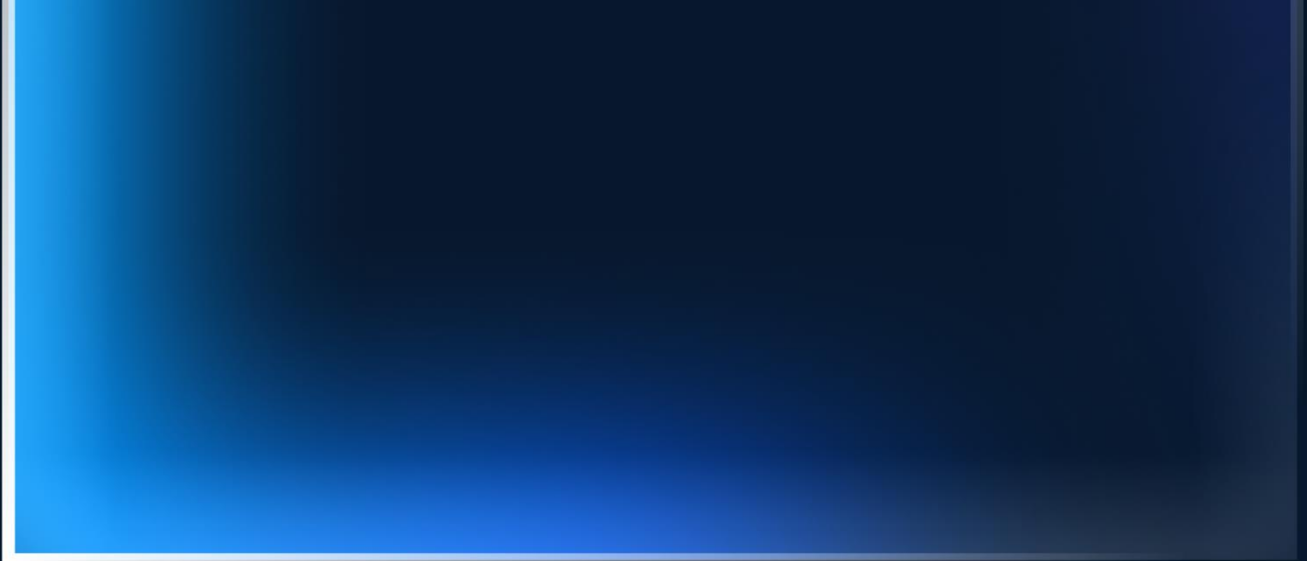
Multicast delivery over a
unicast SRv6 core





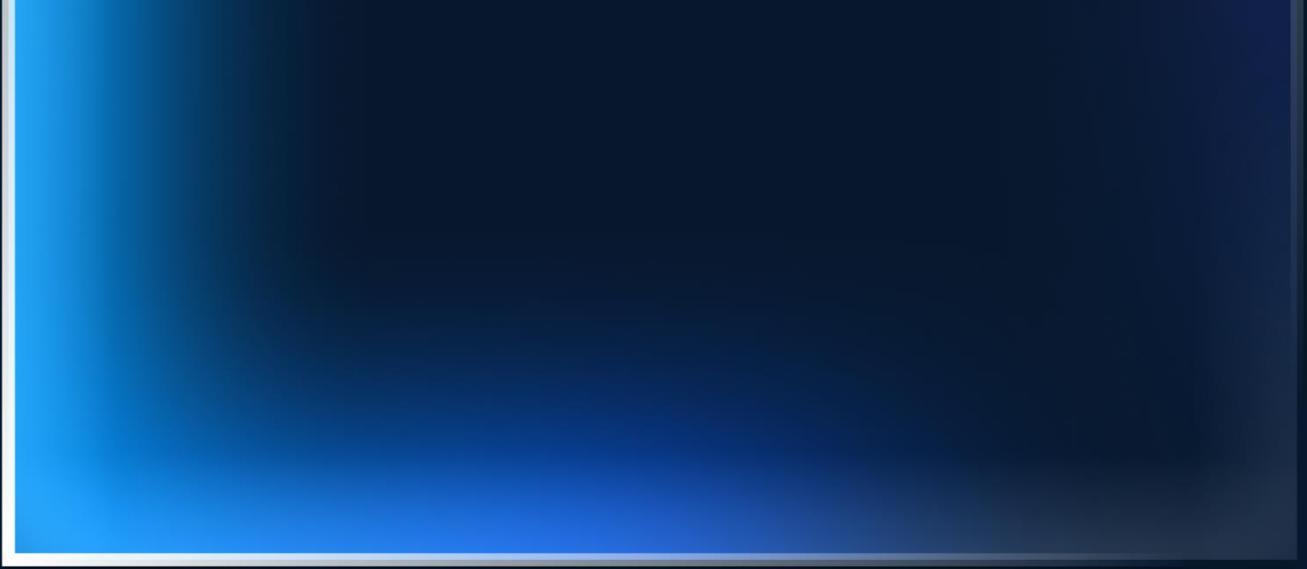
Quantum-Resistant
(Encryption over any transport,
Authentication, Secure Boot)





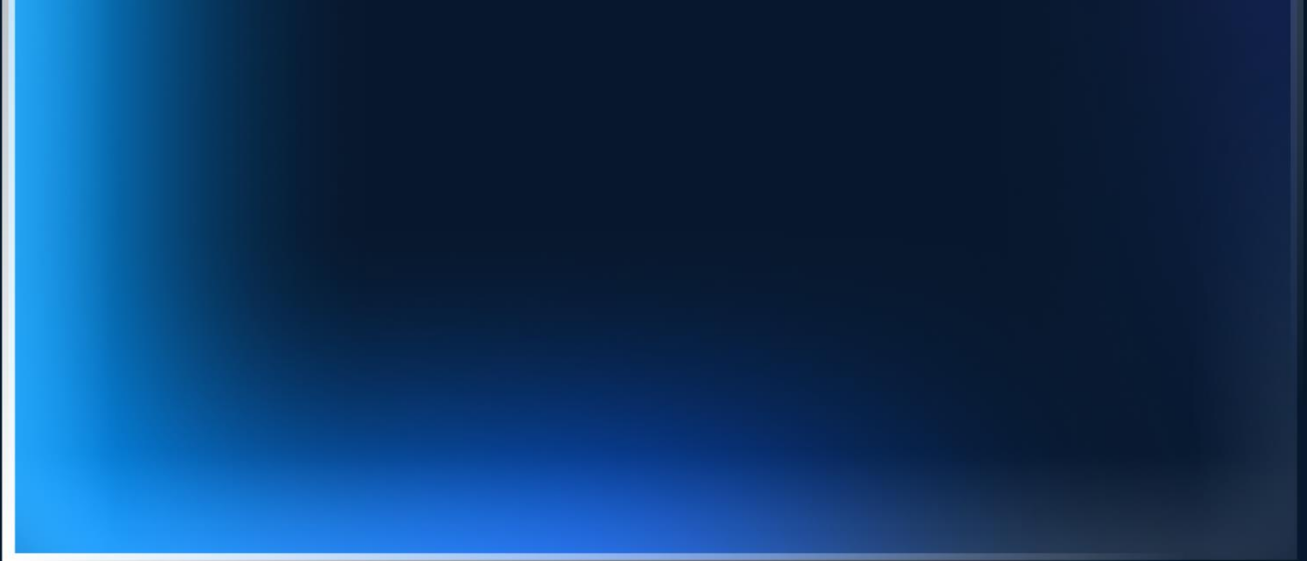
With SRv6 Services





Without ADDING a single,
NEW hardware at the
branch?





That is exactly what I am
going to talk about
TODAY



Introduction

- 26 years with Cisco and 30 total, 2 CCIEs
- Switching, Wireless, SDN Controllers, SDA, SD-WAN, SDA
- “Switched” to Routing last year 📺
- When not working



Kedar Karmarkar

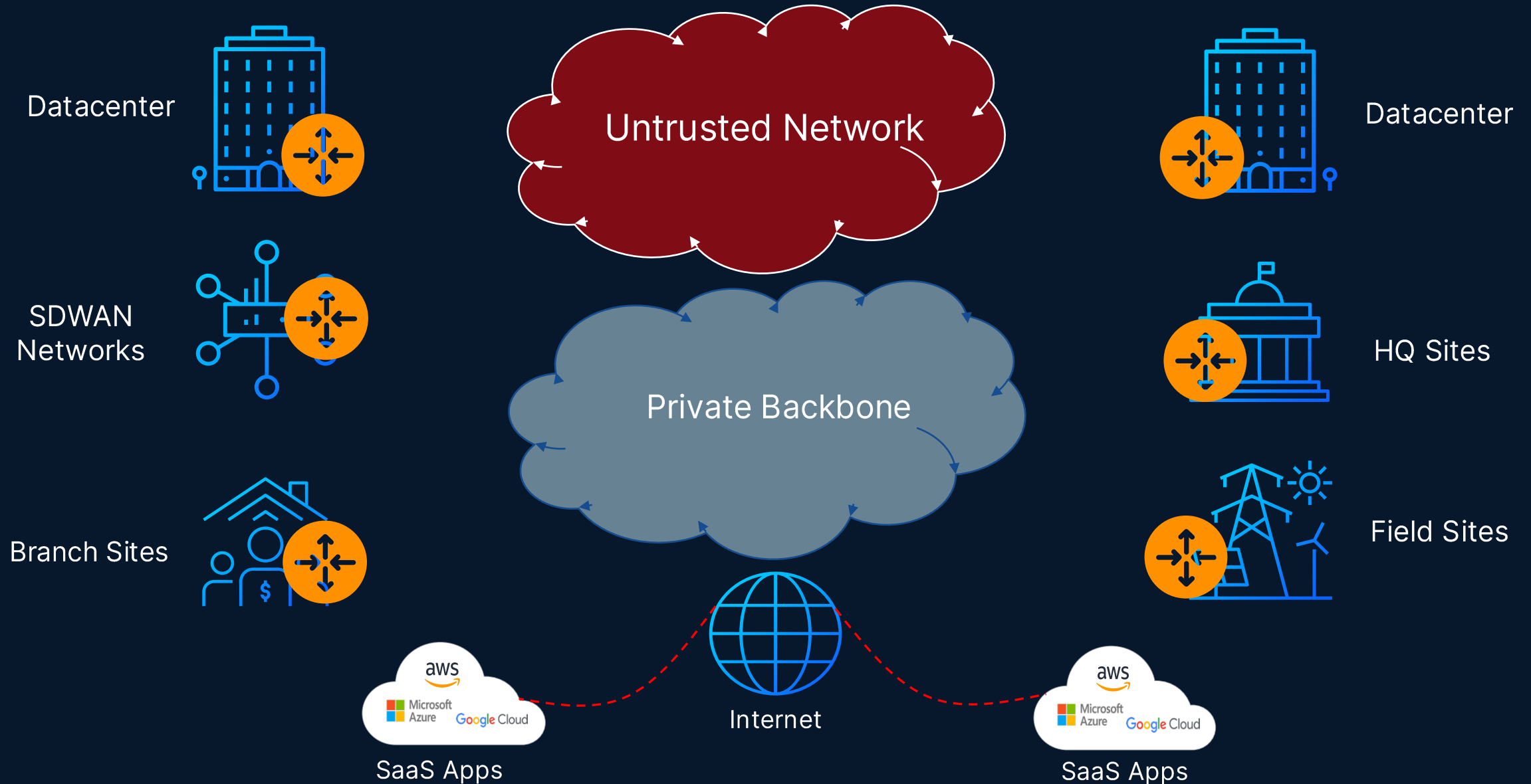
Principal Technical Marketing Engineer

Terminology

“WAN-Edge” “Edge”
“CPE” “Managed CPE” “Access”

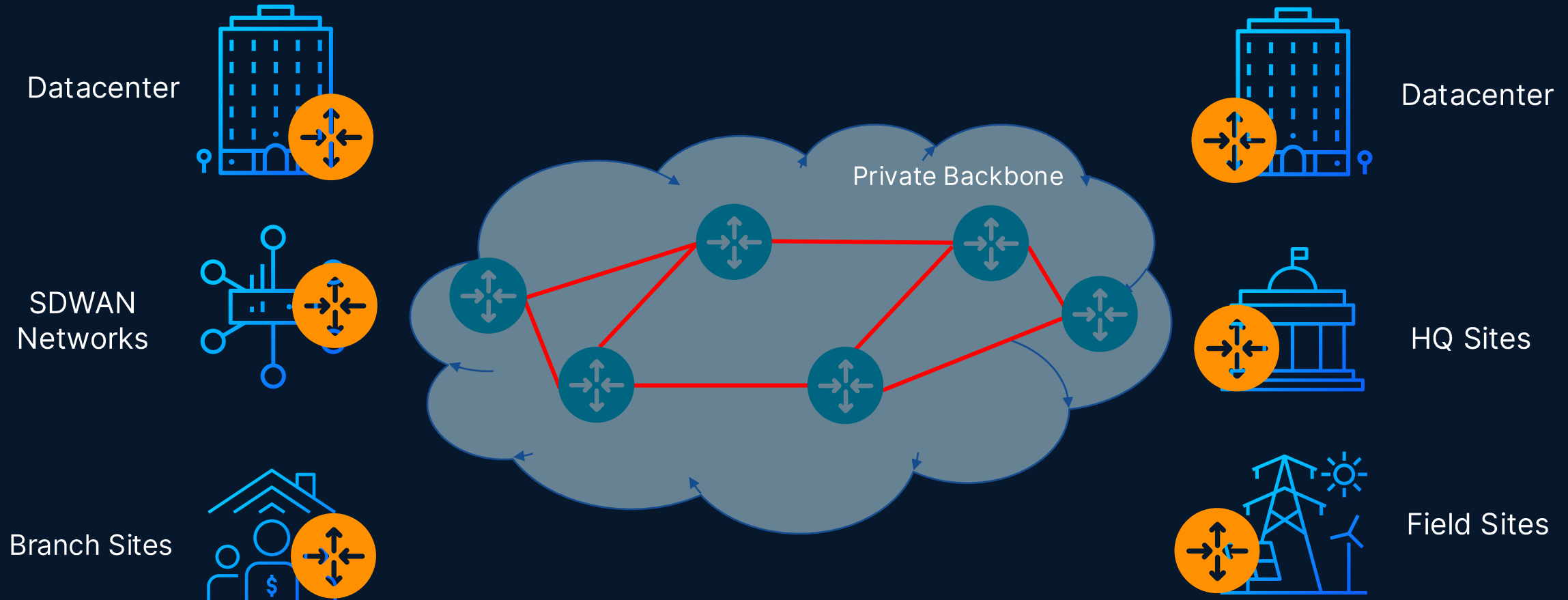
* Encapsulating Router that starts SRv6

Typical Networks Today

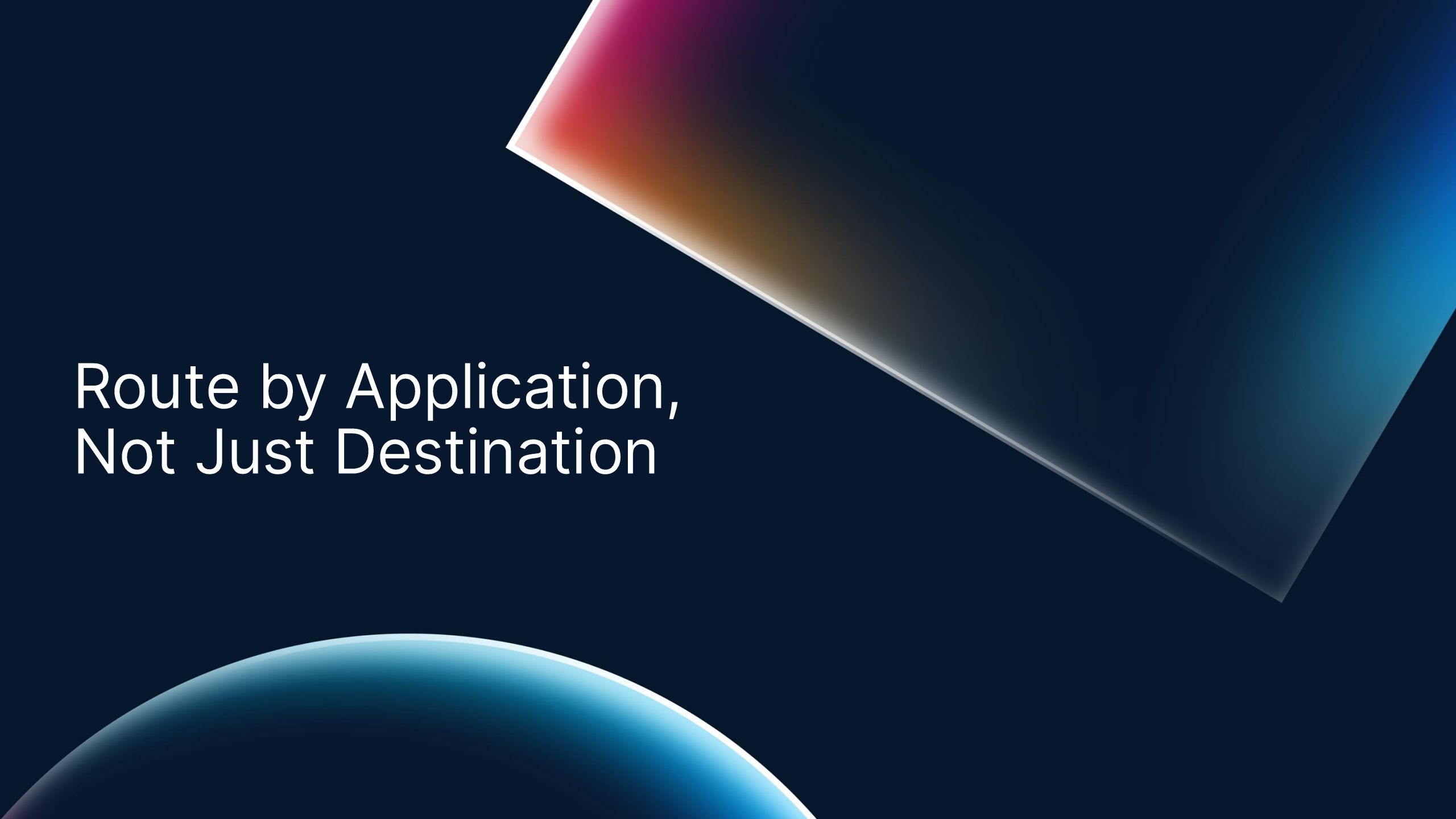


Functional Focus Distribution between Edge and Backbone

Edge focus is on Service integration (Features), Policy Enforcement (Security), Endpoint Connectivity, Connectivity to WAN Fabric

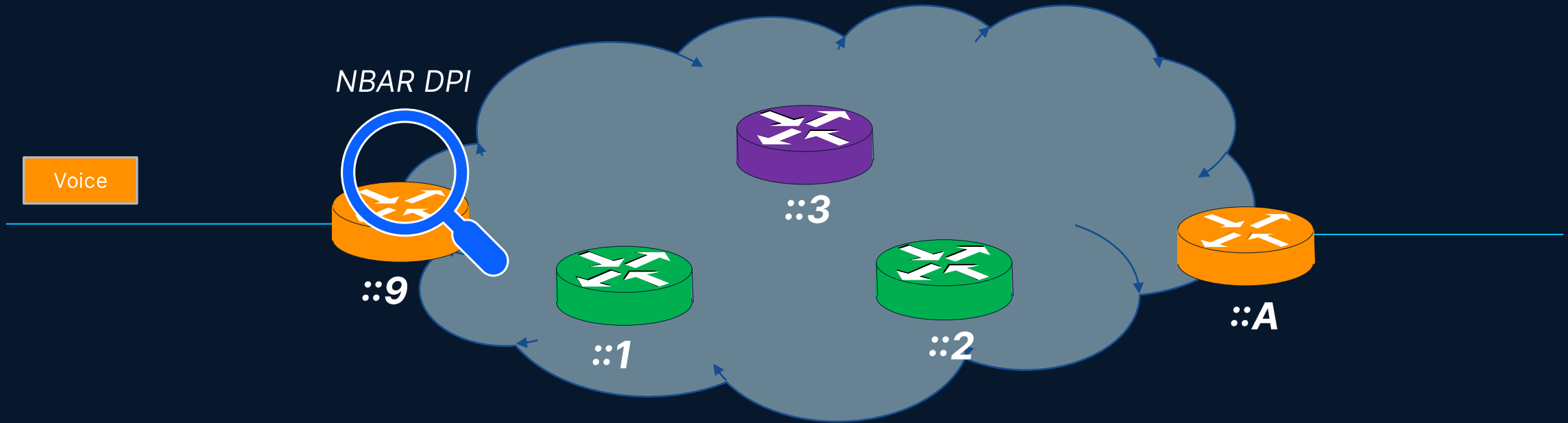


Backbone focus is on High-Speed Switching, Simplicity, Scalable, Redundancy, Fast Failover, ECMP



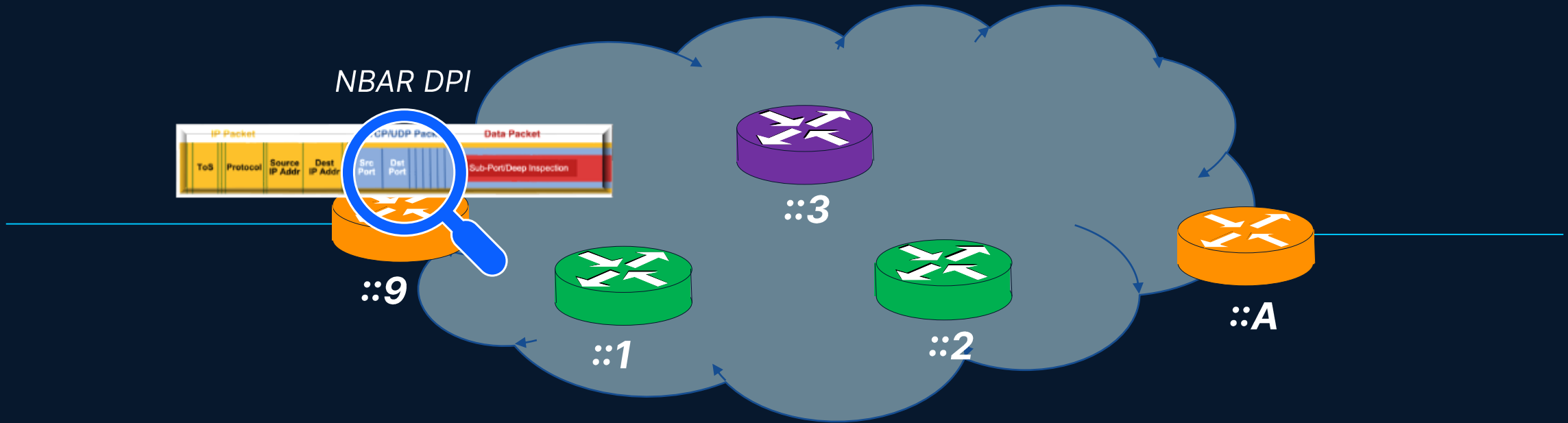
Route by Application,
Not Just Destination

App-Awareness with Network-Based App Recognition



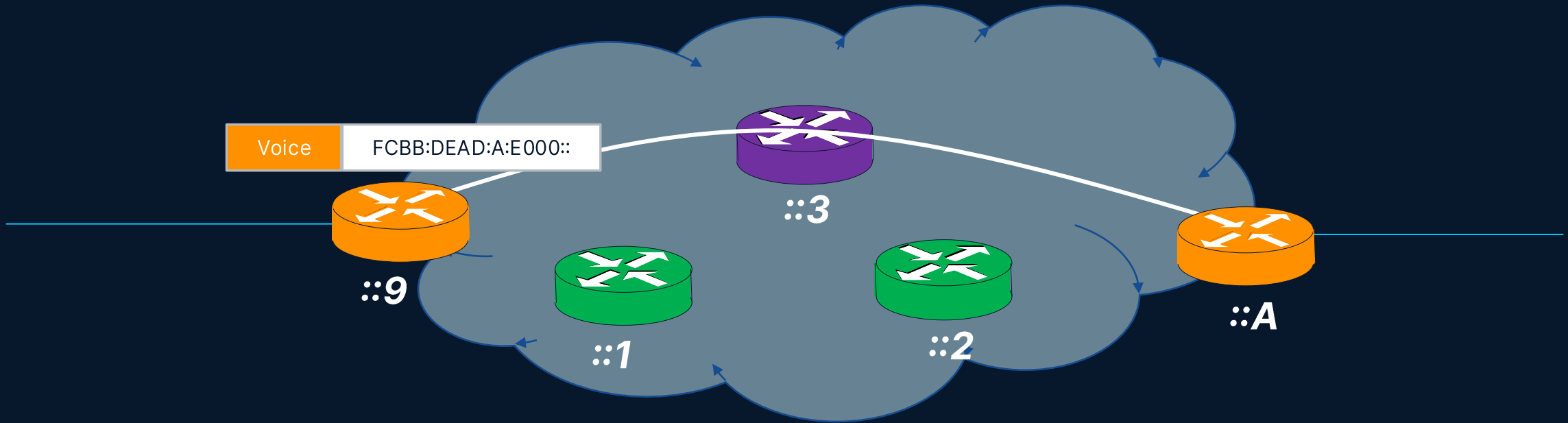
- Application packet enters the router
- Deep packet inspection is done on the packet
- Application is identified by matching the packet against application signatures in the protocol pack.
- First packet match (FPM) is possible via Software-Defined Application Visibility & Control (SD-AVC)

NBAR classification combined with SR-TE PDP/PFP



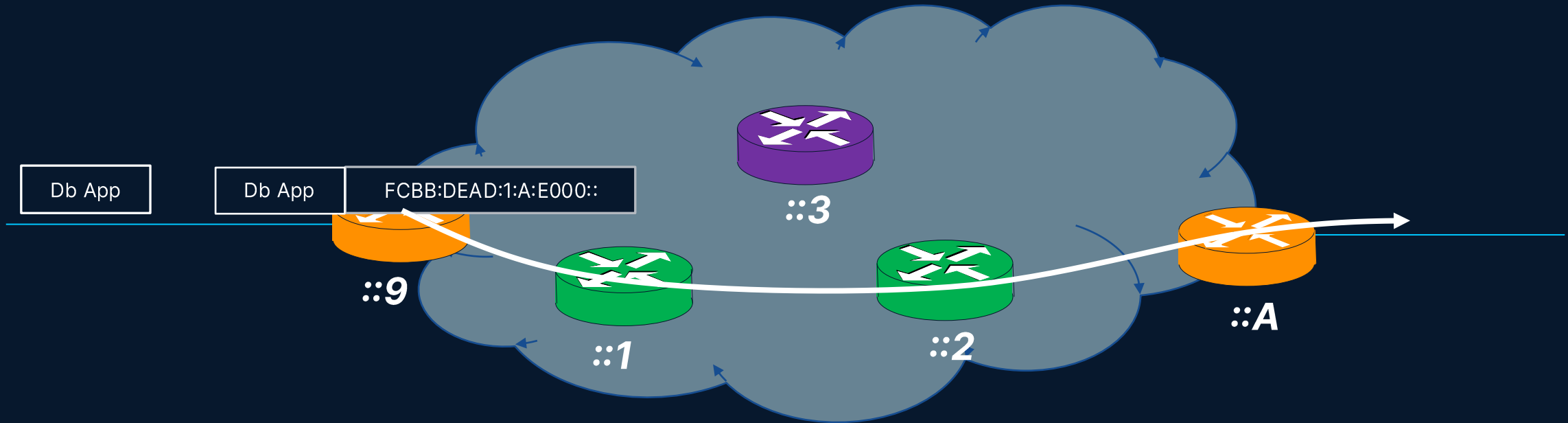
- Application is matched to voice after the matching process
- Forwarding lookup is done on the packet
- Forwarding determines that packet needs to be forwarded via SRv6
- Class-map is mapped to a forwarding class in a Per-Destination Policy within a Per-Flow Policy in SRv6

Voice Application Steered towards Low Delay Path



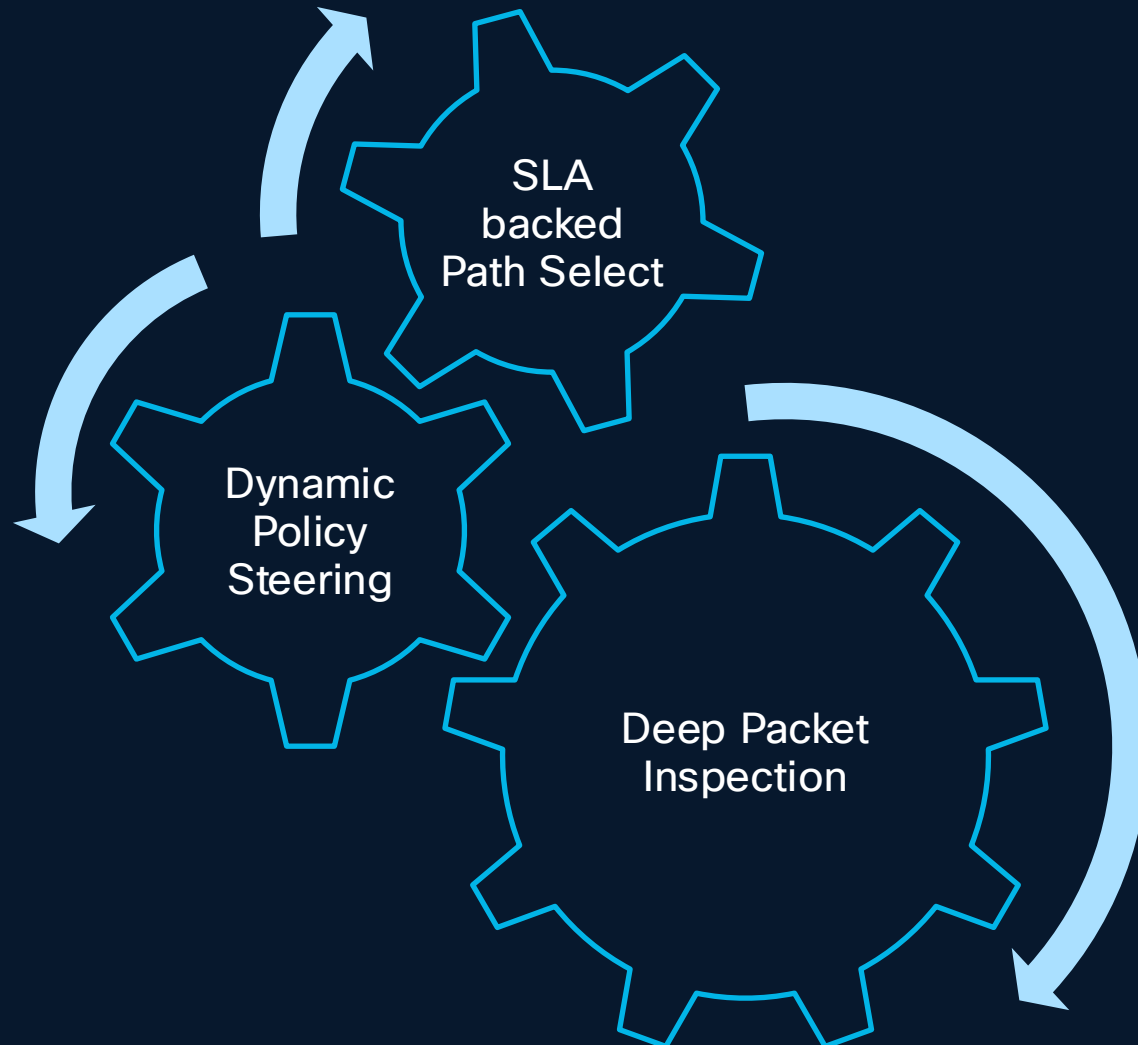
- In this case the packet is mapped to a low-delay SR-TE policy
- Packet is encapsulated in an IPv6 outer-header with the destination address as VPN-SID or a policy SID-List
- And forwarded via the path with the lowest delay

Database Application Steered Towards High Bw Path



- Similarly, a database application packet is identified by DPI and NBAR
- It is mapped to a class-map that is associated with a forwarding class in a PDP/PFP policy
- Packet is encapsulated with IPv6 header where destination address is either VPN-SID or Policy SID-List
- In this case the packet is mapped to a high bandwidth path and forwarded accordingly across the SRv6 core

Route by Application, Not Just Destination



Deep Packet Inspection

via Network-Based Application Recognition (NBAR)

Dynamic Steering

PDP/PFP policy and On-Demand Next-Hop/Automated Steering (ODN/AS)
Combine link-affinity to define network links

SLA-backed

application performance via Path-Selection using IGP metric, delay, Traffic-Engineering (TE)

IOS XE

enhances SRv6 to be application-intelligent



Quantum-Resistant Communications + Product

Building a quantum resistant WAN

Quantum-safe
Communication

Quantum-safe
Product

010110
110010
001011

Encryption

Generate the encryption key for
secure communication



Authentication

Verify the authenticity of the
communicating devices / users



Secure Boot

Ensure authenticity and integrity of
software and hardware

ML-KEM

Key Exchange Mechanism
FIPS-203

ML-DSA

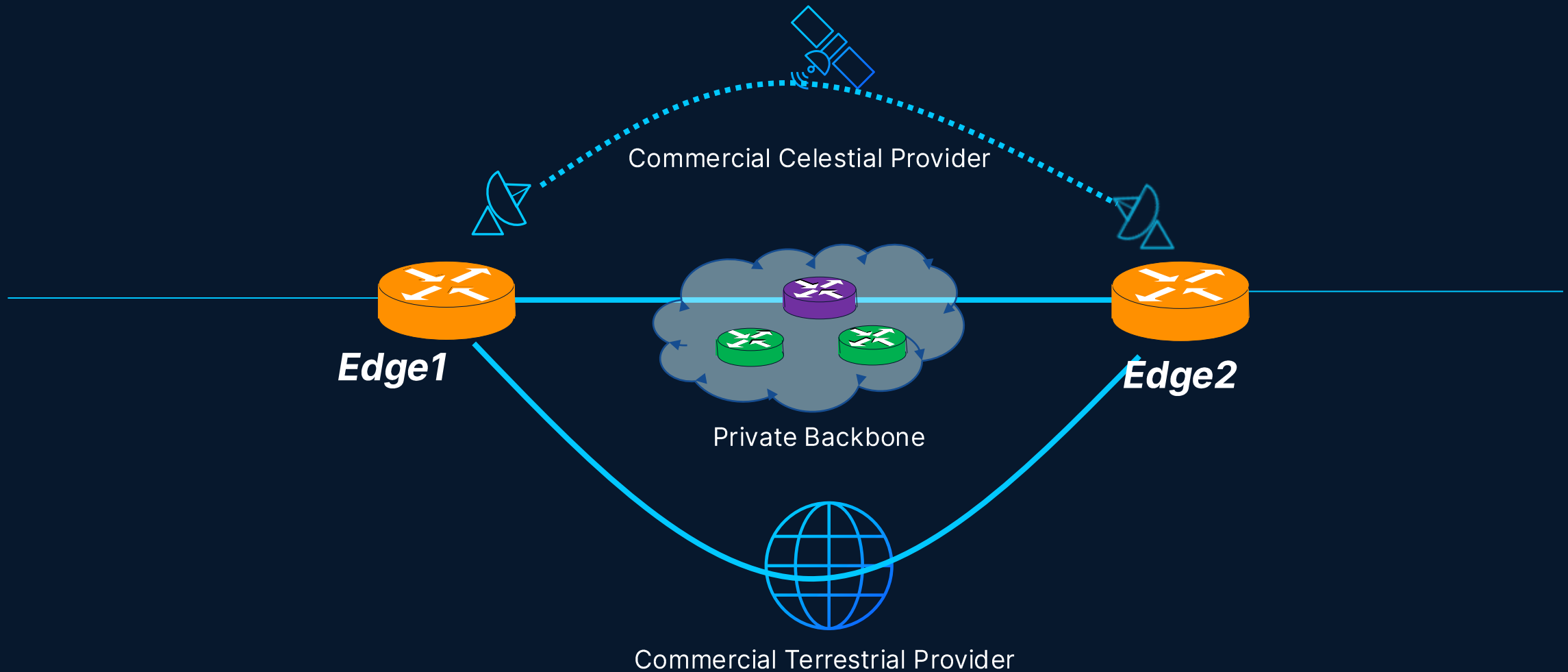
Digital Signature Algorithm
FIPS-204

LMS

Leighton-Micali Signature
NIST SP 800-208

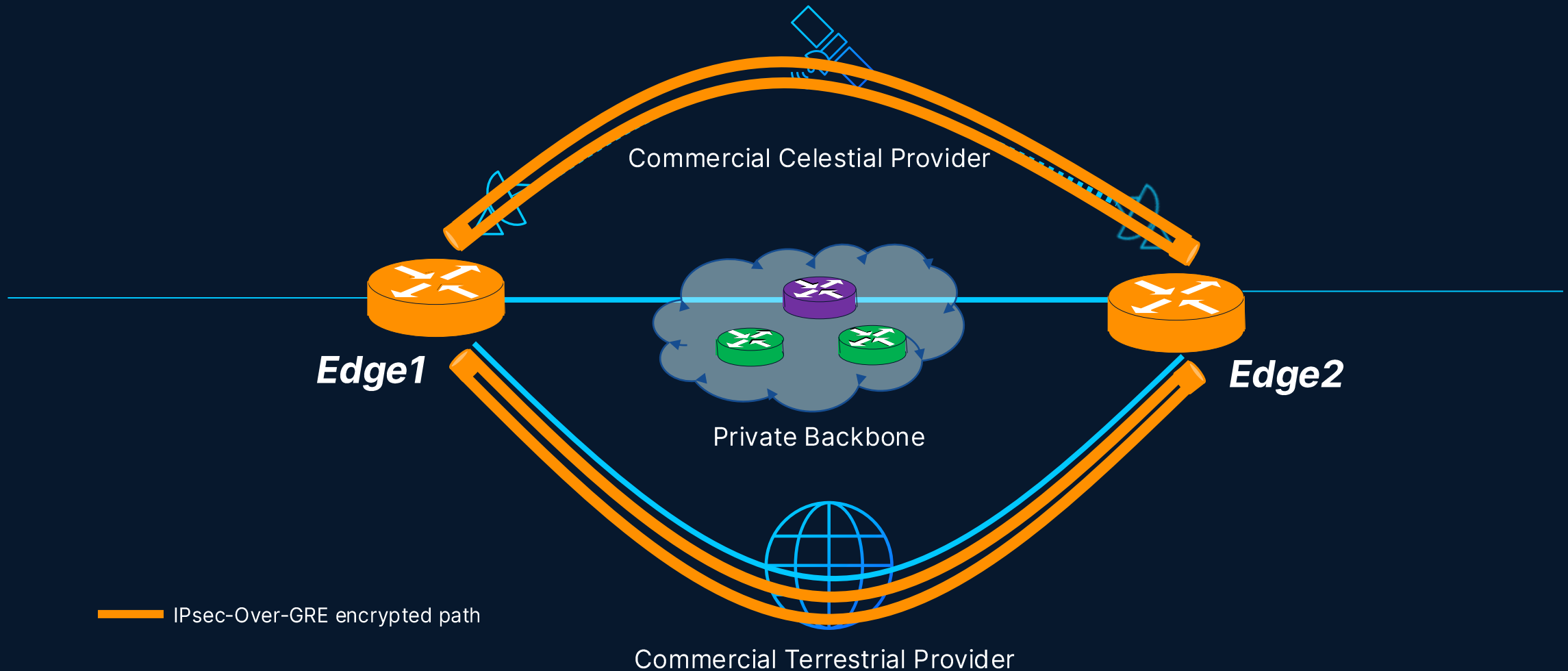
Secure Connectivity Across Untrusted Networks

Multiple Networks for High-Availability



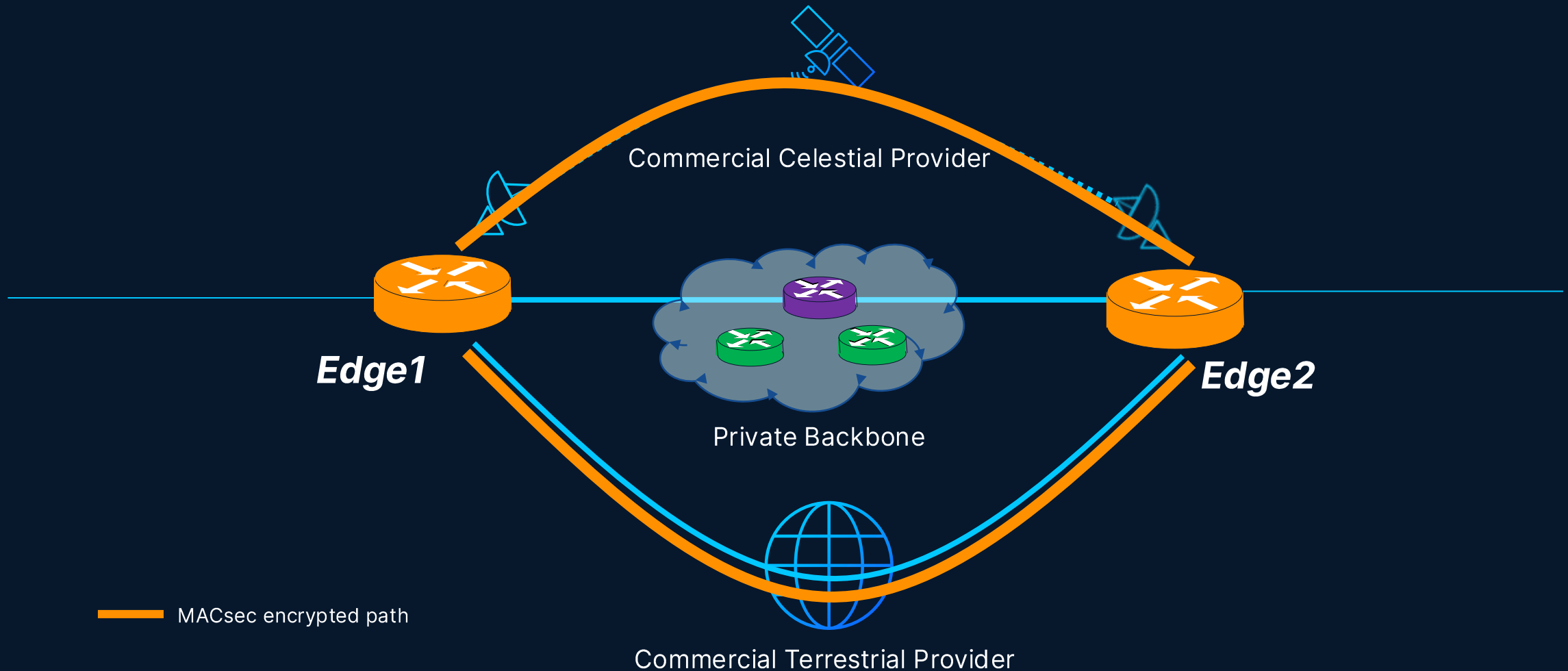
Secure Connectivity Across Untrusted Networks

PQ-Resistant Encryption with IPsec GRE Tunnels in case of L3 connectivity



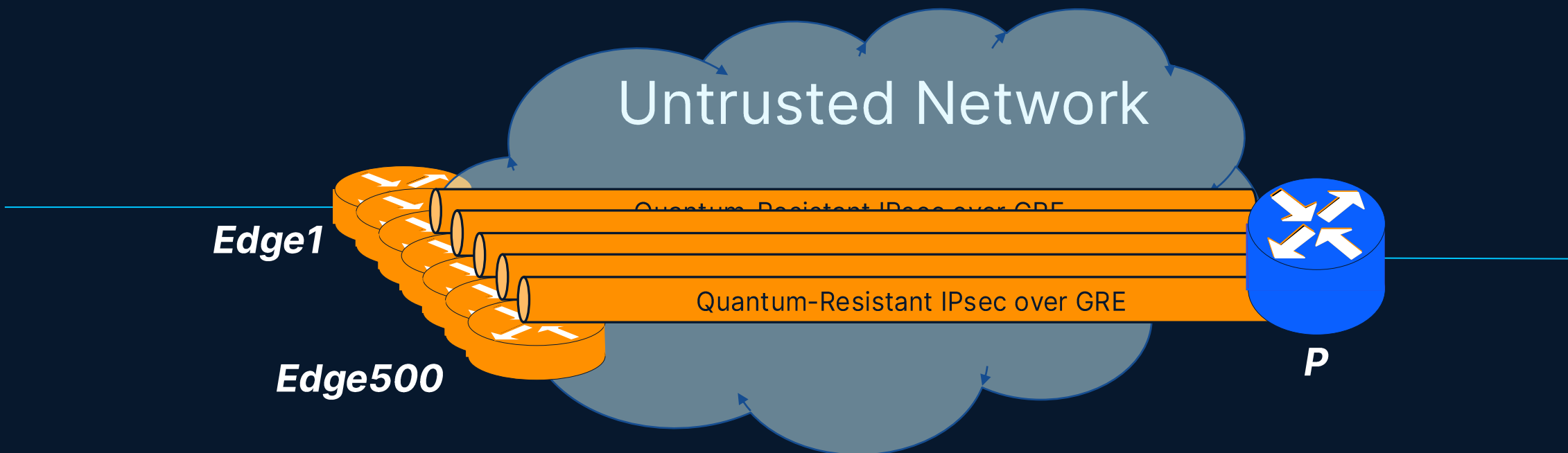
Secure Connectivity Across Untrusted Networks

PQ-Resistant Encryption with Line-rate MACsec in case of L2 connectivity



Secure Connectivity Across Untrusted Networks

PQ-Resistant Encryption Aggregation at Scale with Per-Tunnel QoS



- Up to 500 Q-Resistant IPsec tunnel termination capability at the WAN aggregation/Colo point
- Rich IOS XE features like per Tunnel QoS and many others
- With Cisco 8500/8600 Series Secure Routers

Securing the Untrusted Transport

Key Capabilities

Quantum-resistant Key Exchange/Establishment ensures data integrity against HNDL attacks

Quantum-Resistant Encryption at Scale with Per-Tunnel QoS Services

Line-rate Quantum-resistant WAN MACsec

Quantum-Resistant Communications (Available Today) + Product (Available by EoY)

Strategic Takeaway

IOS XE secures SRv6 against both today's threats and tomorrow's quantum computers



Force Multiplier

Centralized Internet Gateway

Centralized Internet: Centralized Services

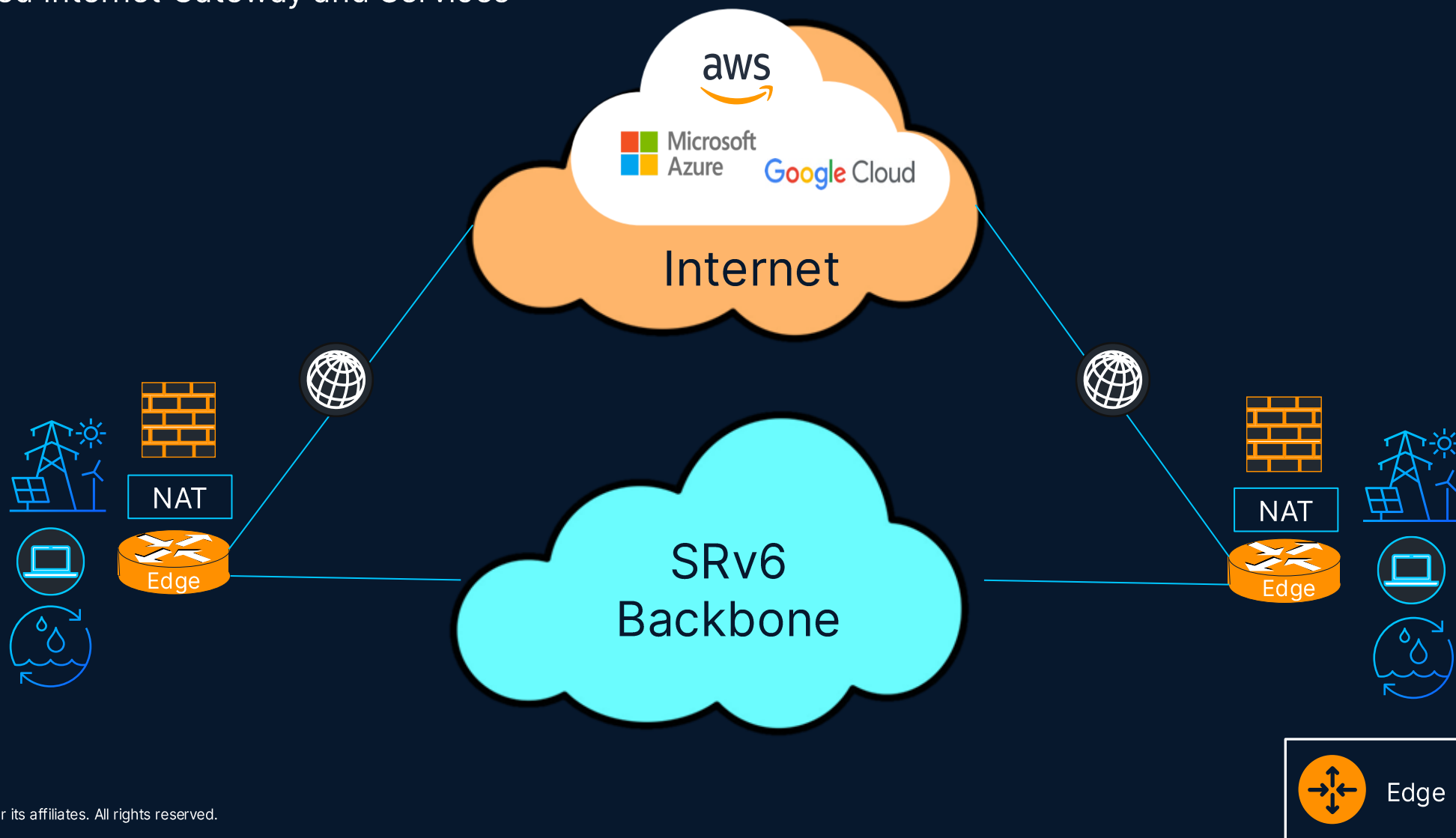


Centralized Internet: Distributed + Centralized Services



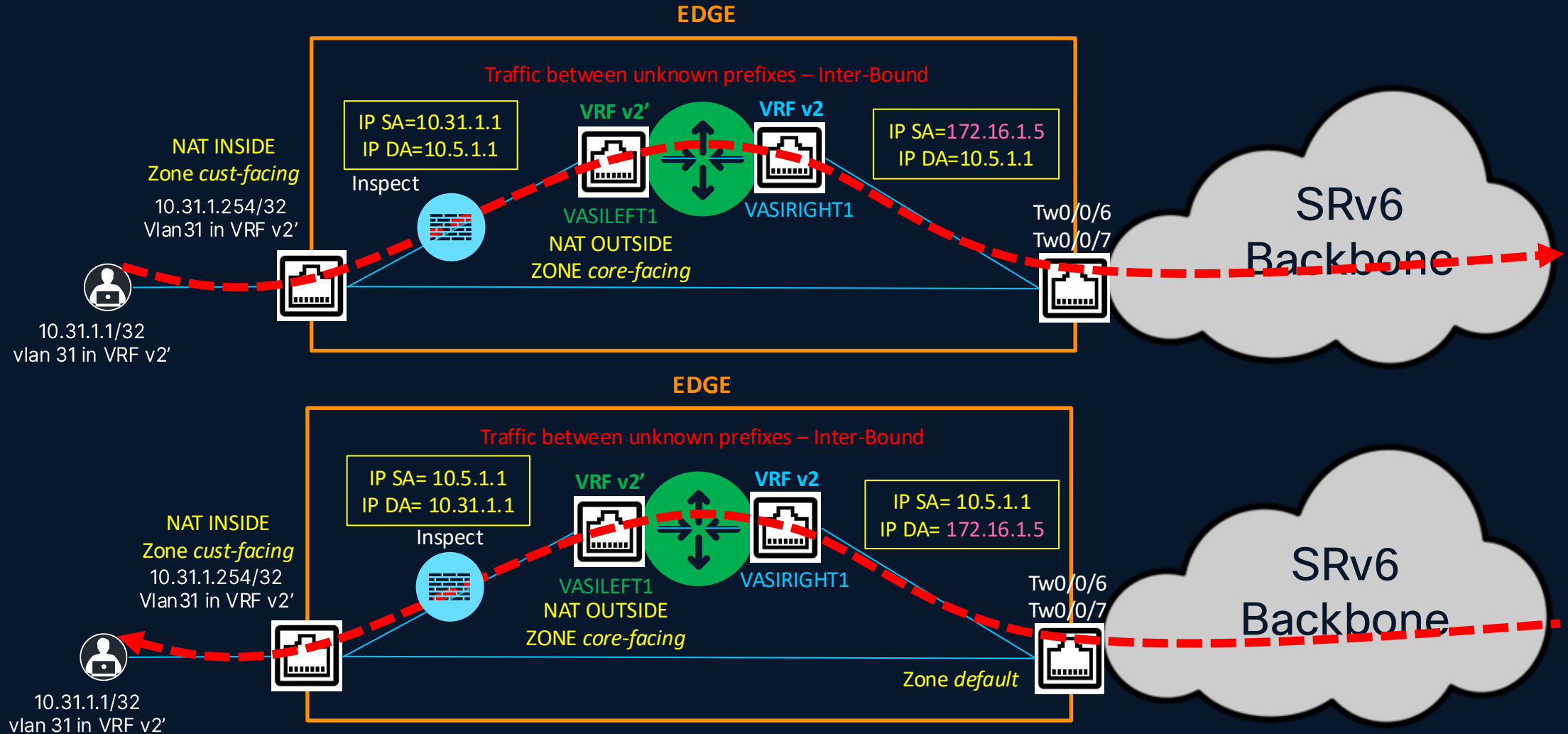
Direct Internet Access (DIA) with L3VPN across SRv6

Distributed Internet Gateway and Services



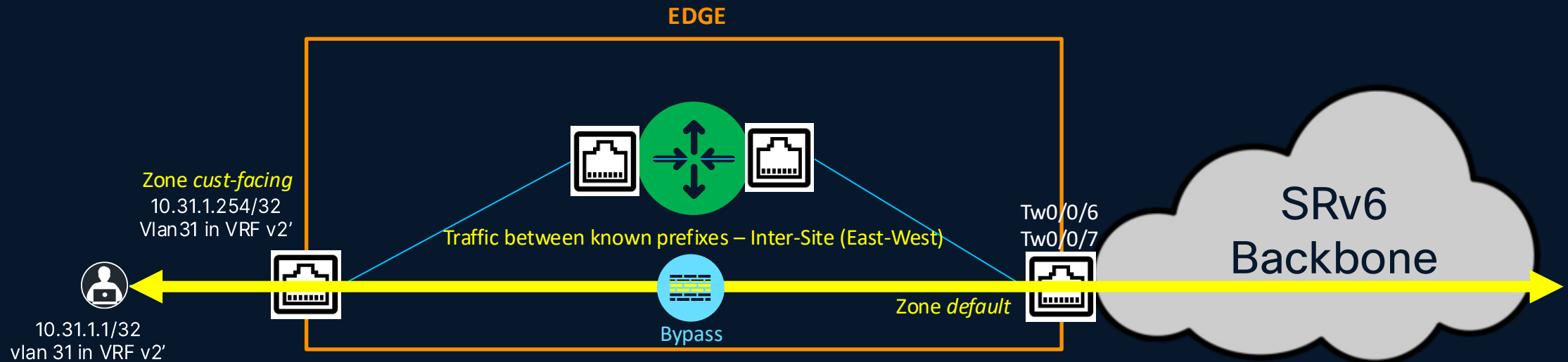
Selective Traffic Re-direction for Internet traffic

Centralized Internet: Distributed Services



Selective Traffic Re-direction for Intranet traffic

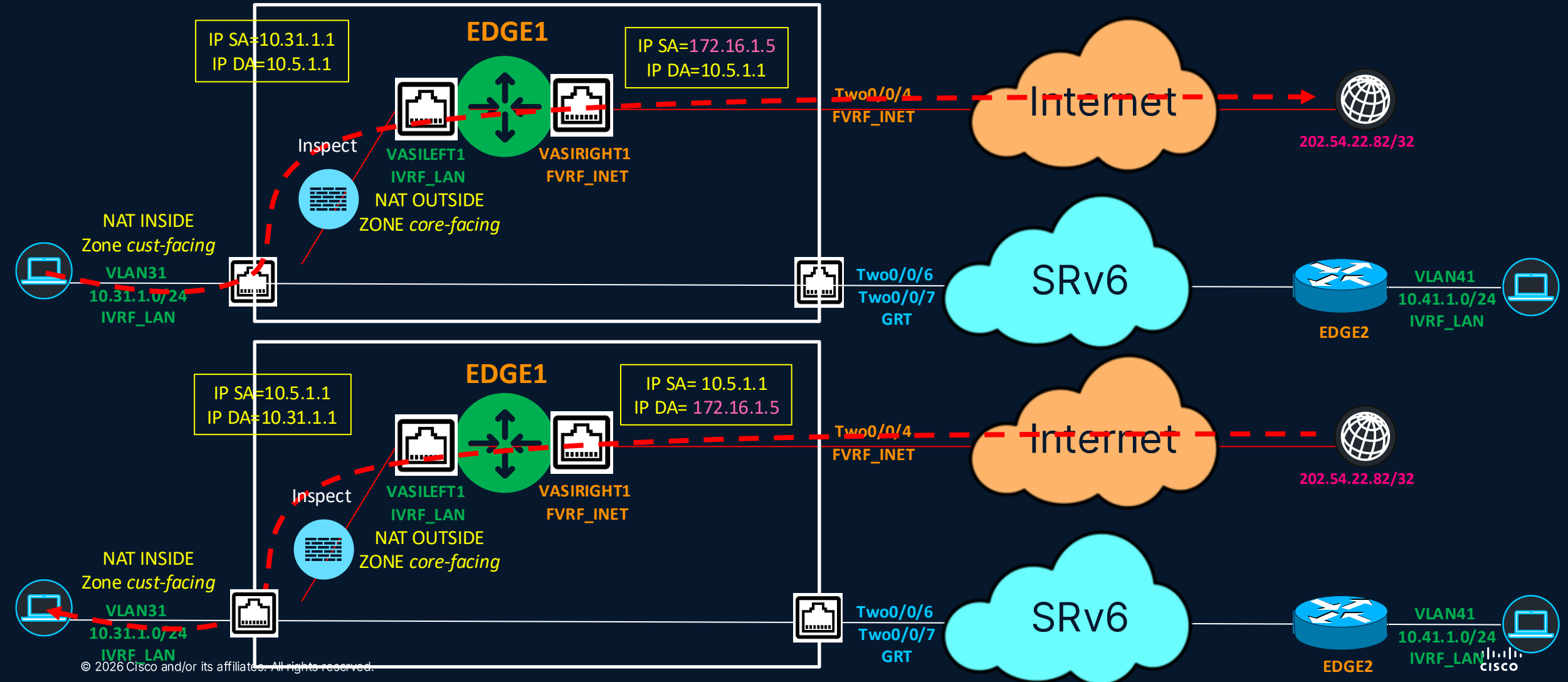
Centralized Internet: Services Bypassed



<https://www.cisco.com/c/en/us/solutions/collateral/design-zone/cisco-validated-profiles/central-internet-gateway-with-distributed-network-services-wp.html>

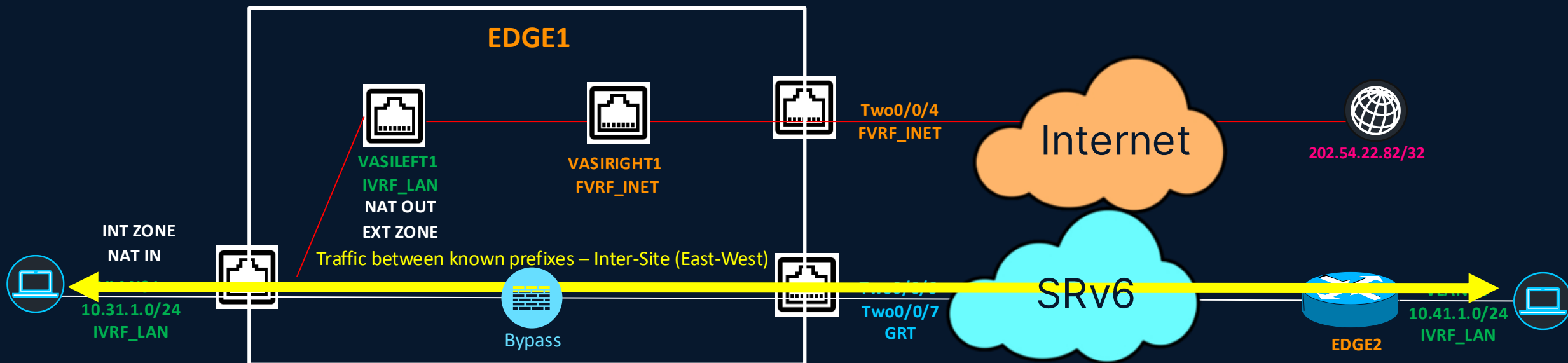
Selective Traffic Re-direction for Internet traffic

Distributed Internet: Distributed Services



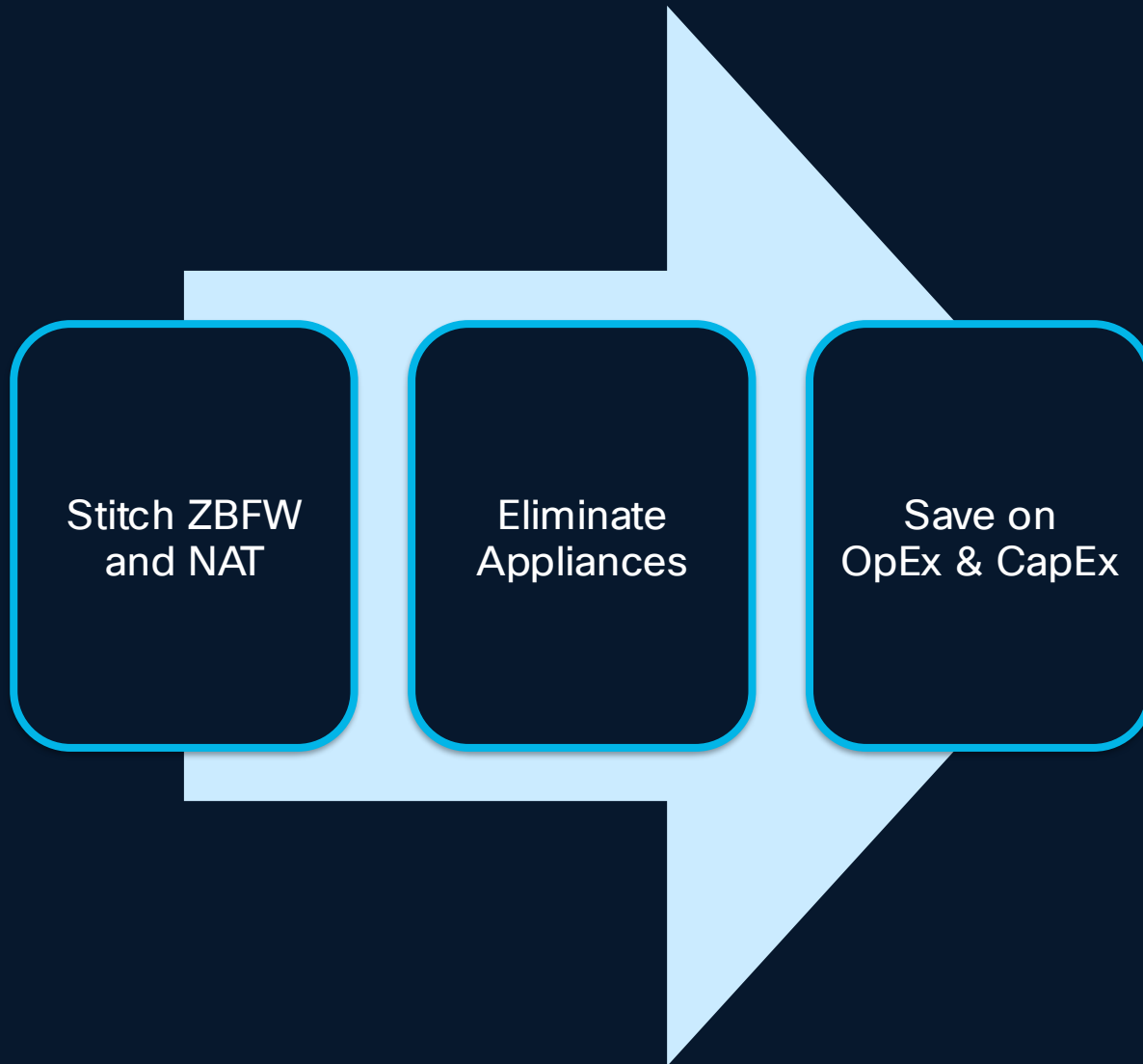
Selective Traffic Re-direction for Internet traffic

Distributed Internet: Services Bypassed



<https://www.cisco.com/c/en/us/solutions/collateral/design-zone/cisco-validated-profiles/local-internet-breakout-over-networks-wp.html>

Force Multiplier – Service Consolidation



VRF-Aware Software Infrastructure (VASI)

Stitch NAT and Firewall directly into SRv6 CPE

SPs/MSPs can monetize incremental services

One Router = Many Services = Force Multiplier

Cisco 8000 Series Secure Routers

Force-Multiplier



**Cisco 8100 Series
Secure Routers**

Small Branch



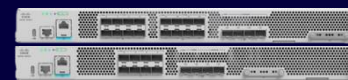
**Cisco 8200 Series
Secure Routers**

Medium Branch



**Cisco 8300 Series
Secure Routers**

Large Branch



**Cisco 8400
Series Secure
Routers**

Campus



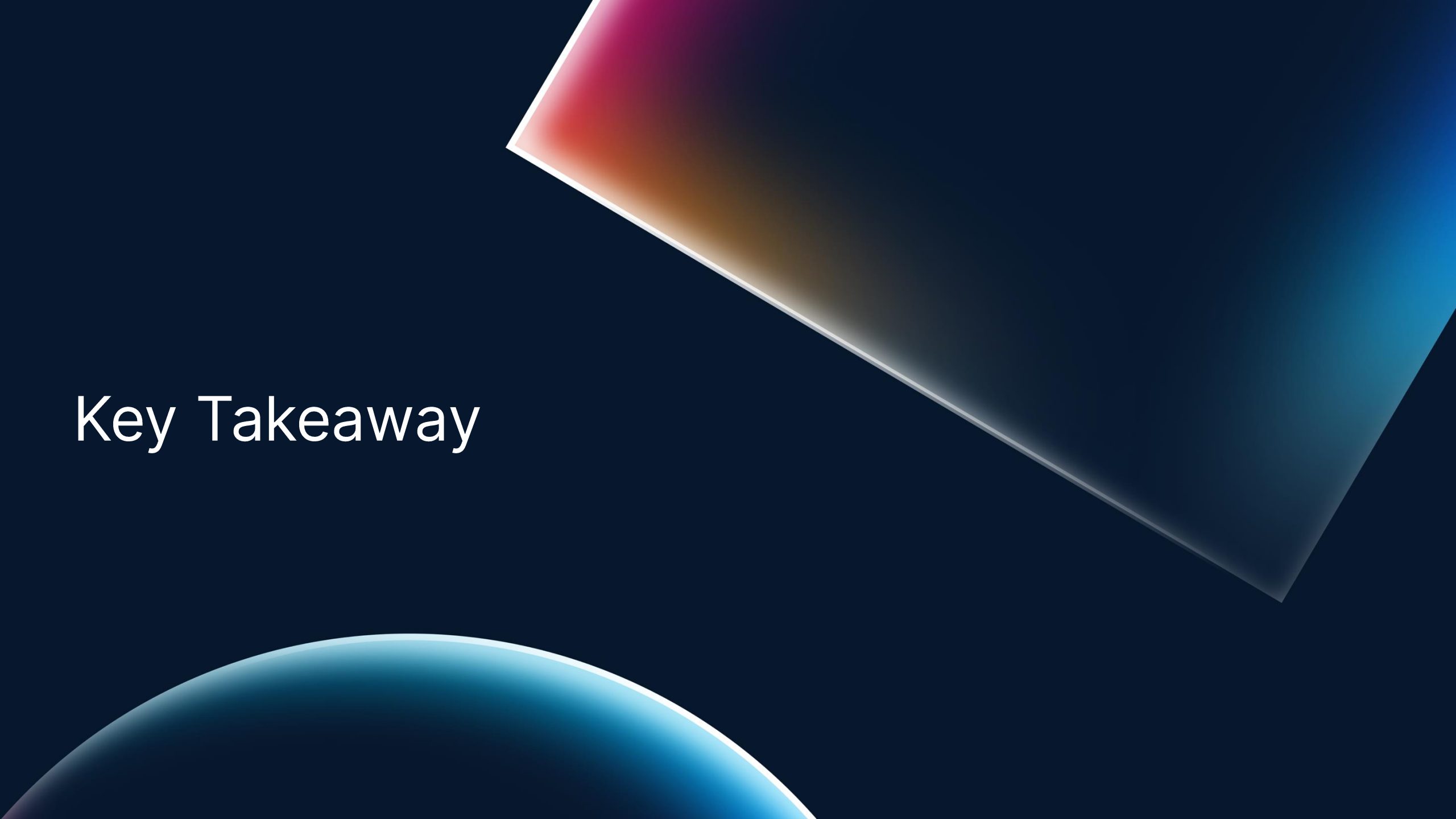
**Cisco 8500, 8600
Series Secure
Routers**

Data Center

**App-Aware Routing, Multicast over Unicast, Firewall, NAT,
Quantum-Safe Encryption at Scale with SRv6 Services**

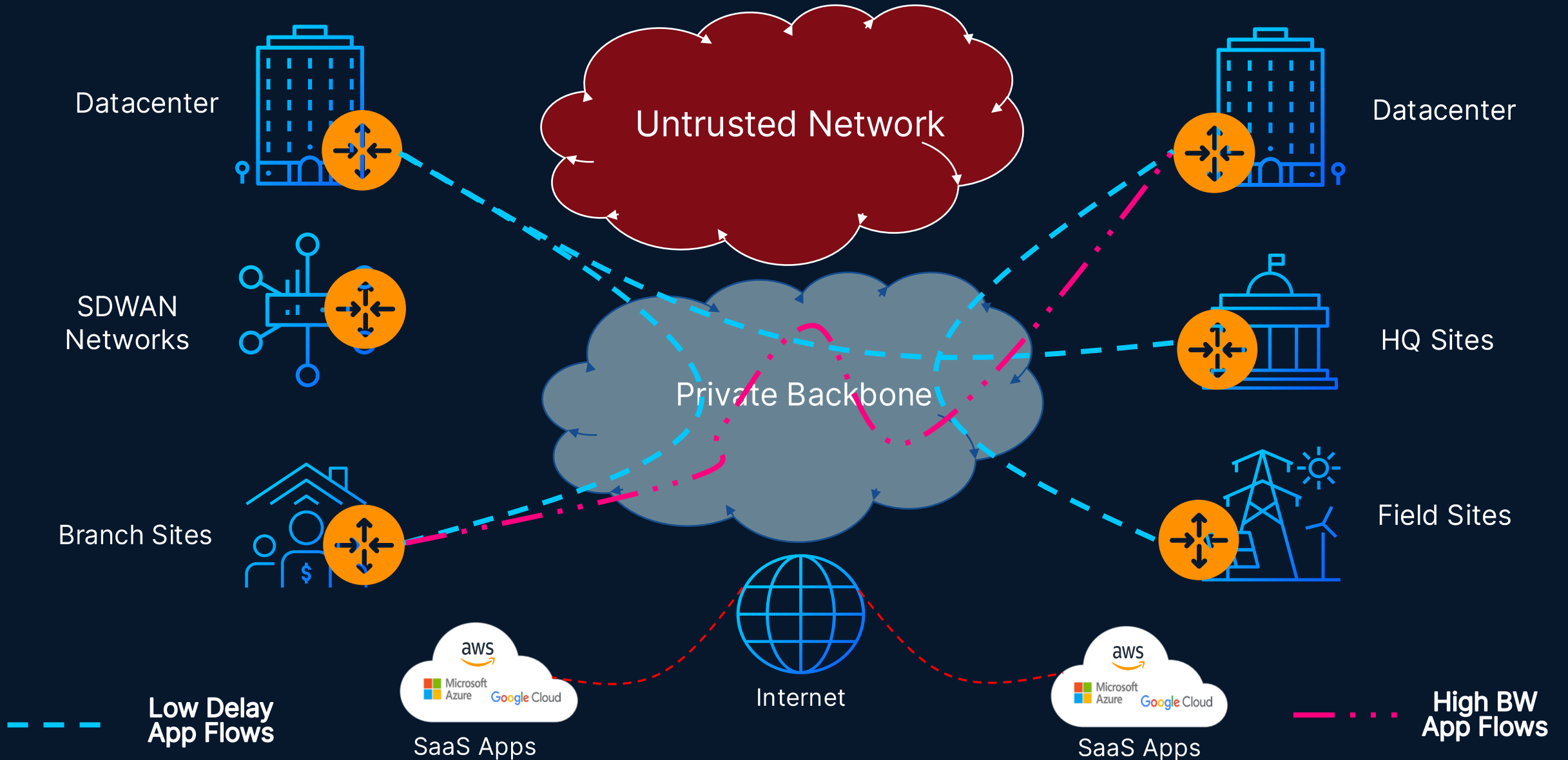
Links to CVD, White Papers

- Enhance SRv6 with Application-Aware Routing -
<https://www.cisco.com/c/en/us/solutions/collateral/design-zone/cisco-validated-profiles/CVD-Application-Aware-Routing-in-SRv6.html>
- Direct Internet Access in SRv6 networks –
<https://www.cisco.com/c/en/us/solutions/collateral/design-zone/cisco-validated-profiles/local-internet-breakout-over-networks-wp.html>
- Centralized Internet Gateway with Distributed FW/NAT –
<https://www.cisco.com/c/en/us/solutions/collateral/design-zone/cisco-validated-profiles/central-internet-gateway-with-distributed-network-services-wp.html>
- Multicast Delivery over a unicast SRv6 Core -
<https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/mvpn-ir-over-SRv6-publish-wp.html>
- Building Quantum-Resistant Mission-Critical Networks with SRv6 –
https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/Building_Quantum-Resilient_Mission-Critical_Networks_with_SRv6.html

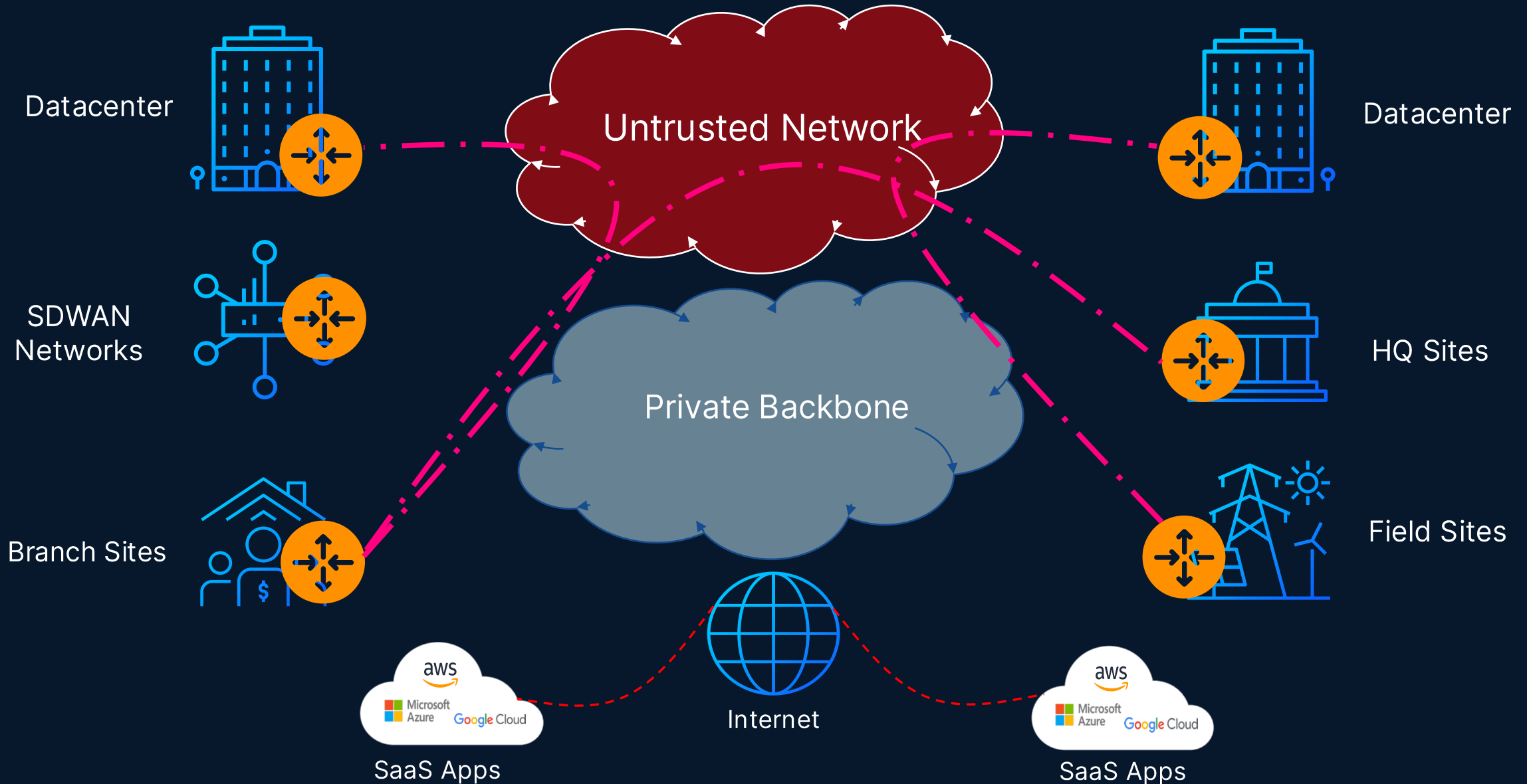


Key Takeaway

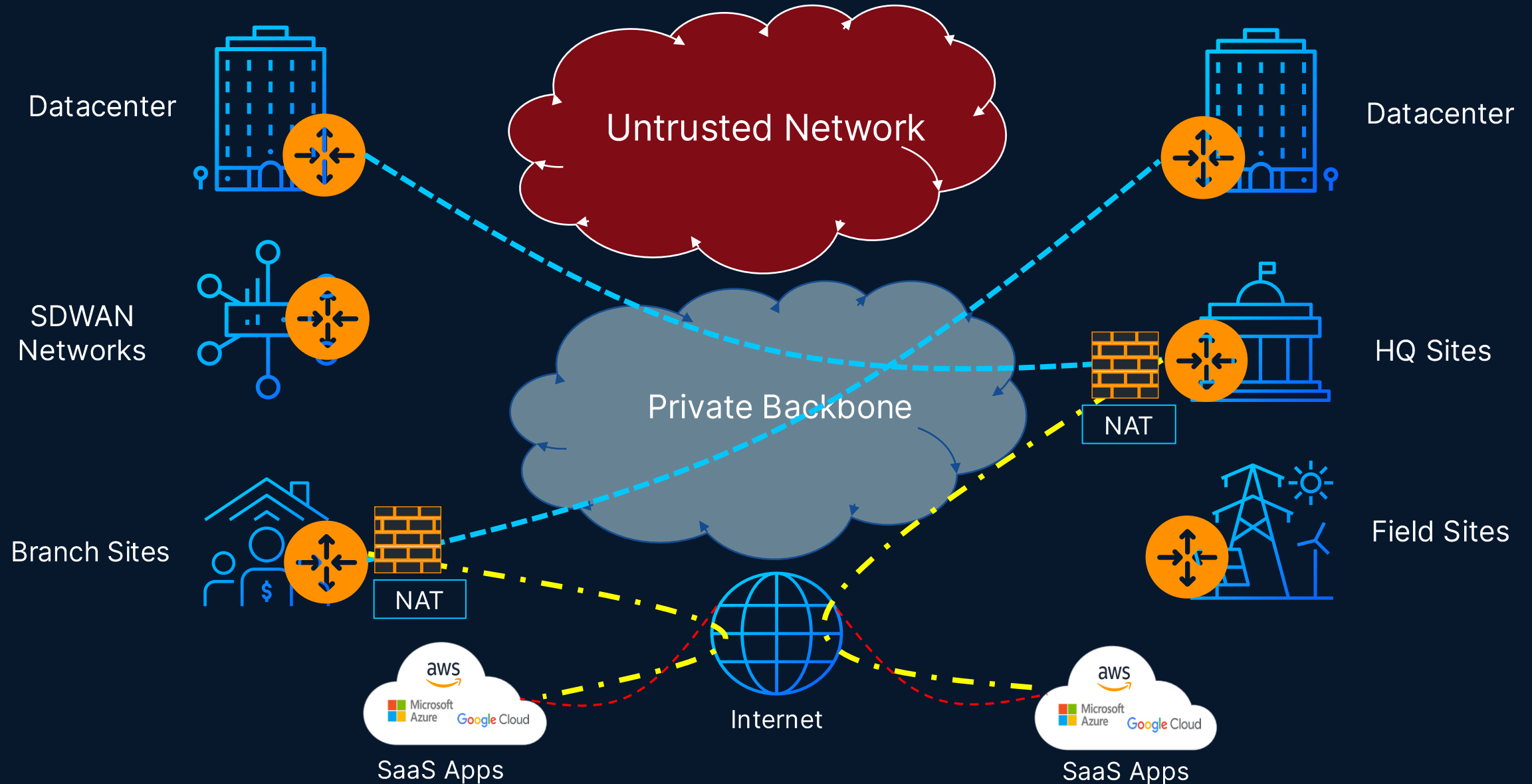
Application-Aware Intelligence to SRv6 Steering



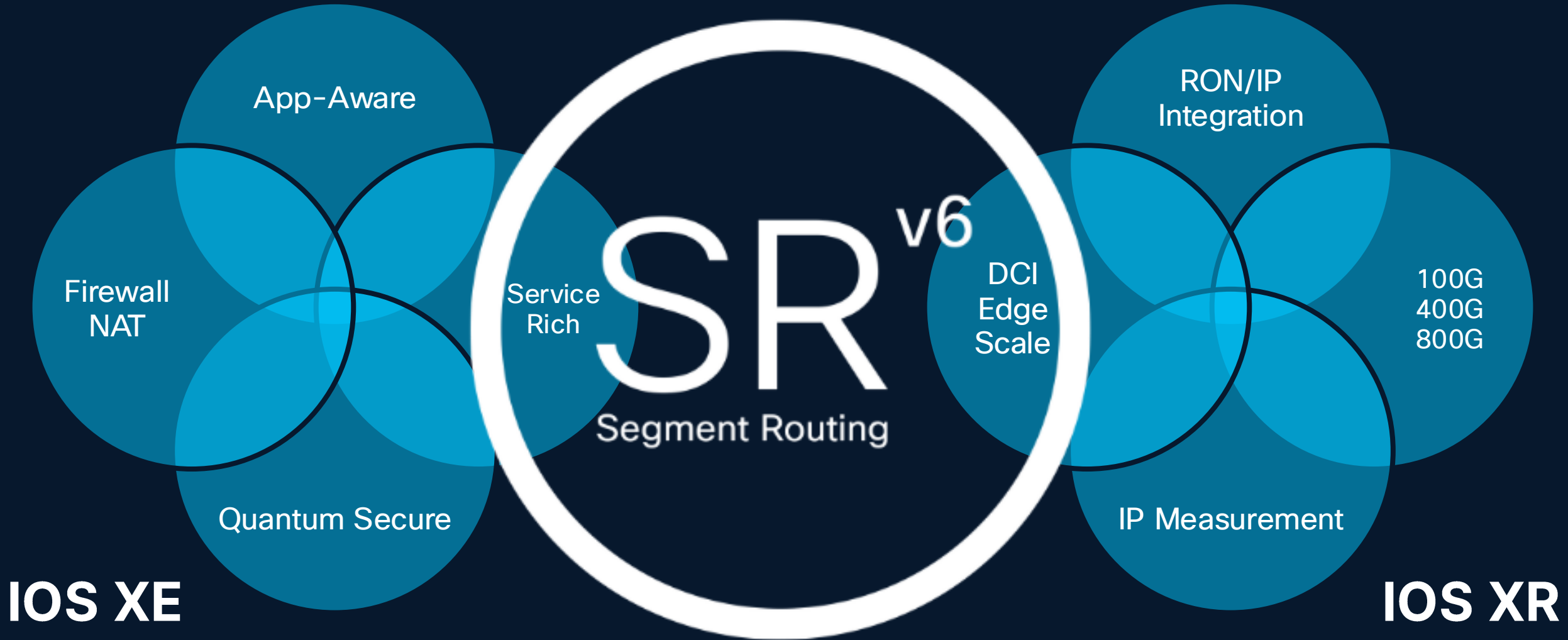
Quantum-Resistant Communications + Product



Distributed FW+NAT for Selective Internet-bound Traffic



Value-added Services with Simplicity of SRv6



Cisco: Making SRv6 Great ... Since 2016

CISCO Live !

Thank you



